AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 18 · WORKBOOK

Confidence Is Not Correctness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected confidence audit.

Interpret a match score, judge case outcomes, compare claimed and observed fractions and reject duplicated evidence.

Keep one habit: Keep the claim, checked result and unknown cases visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Prediction Is Not Fact

Save New Examples for Testing

Debug the Cause

W1 · Interpret a matching score honestly

> Use its documented feature-point meaning.

Your evidence record

- A card sorter gives a match score of 80 out of 100 points. Its documentation says the score counts weighted feature matches and is not a probability. A learner calls this an 80 percent chance of **correctness**.

  Explain the error and write an accurate description.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Judge correctness from outcomes

> Use the supplied reference cases and retain the unassessed case.

Your evidence record

- Reference cases: A predicts far with a stated 90 percent probability; the verified outcome is near. B predicts near with a stated 60 percent probability; the verified outcome is near. C predicts far with a stated 90 percent probability; its outcome is unavailable. These are fictional claims and checking records, not validated probability estimates.

- Use cases A, B and C from the lesson table. State which is correct, incorrect and unassessed. Explain why ordering them by **confidence** cannot replace checking their outcomes.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Compare a claim with observed frequency

> Keep the reported probability and five-of-ten result separate.

Your evidence record

- Ten **prediction**s each claim an 80 percent probability that their chosen category is correct. Five later prove correct and five incorrect. State the claimed probability and observed fraction separately.

  Does this prove every future **prediction** will be wrong, or that each next **prediction** has exactly a 50 percent chance? Explain.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Reject duplicated predictions as new evidence

> Request actual new relevant outcomes rather than more copies of a claim.

Your evidence record

- A learner copies the same 90 percent **prediction** into ten rows and calls this ten independent tests. No new outcomes were obtained. Explain why the **evidence** has not grown and propose a useful check.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit a different probability claim

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____
>
> _____

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Audit a different probability claim

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Record both the sorter's **confidence** wording and the human-confirmed outcome; keep them in separate columns.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Judge cases using outcomes for **Confidence** and **correctness**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Confidence** and **correctness**. Proposed explanation: “An 80 point score always means an 80 percent probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A helper confidently labels a blurred dog photo as cat; the clear original shows a dog.

What does this show?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorter is very confident that a leaf is a coin, but a person checks and sees a leaf. Which determines **correctness**: **confidence** or the checked object? Why?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 18 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A helper says 'definitely correct' on four answers; only two match the answer key. A cautious helper gives the same four answers. Which is more accurate? What additional **evidence** would you need to compare **confidence** quality?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional leaf sorter claims a 75 percent probability of **correctness** for each of eight chosen labels. Later **evidence** establishes six correct labels, one incorrect label and one missing outcome.

A report says six out of eight proves perfect **calibration** and guarantees the next label. Repair the report, state the assessed and unassessed counts, and describe a better follow-up check.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

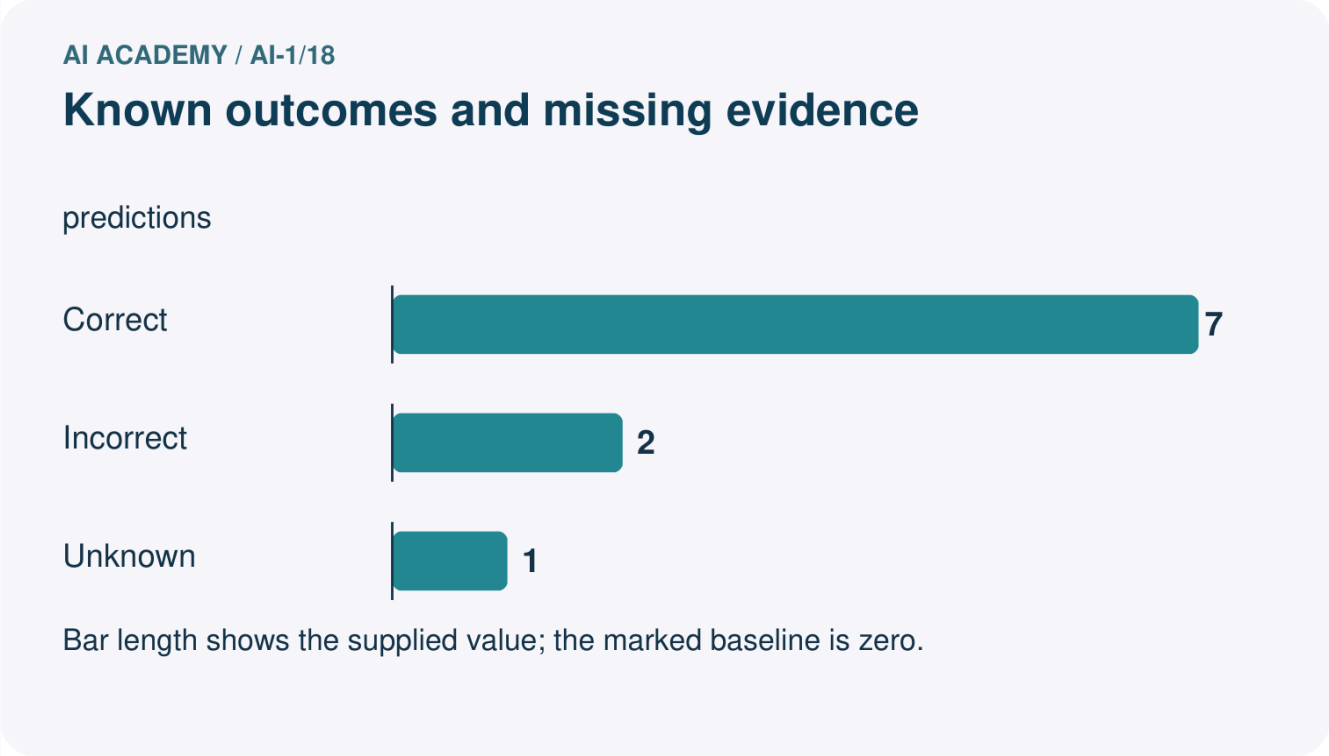
> Keep the explanation and the support you used.

Use this when helpful

A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.

Ten **prediction**s all have score **80%**.

Later checks find 7 correct, 2 incorrect and 1 outcome unavailable. The selected cards were not a representative survey.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Was the unknown label guaranteed correct because its **confidence** was **80%**?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional card classifier reports a claimed **80%** chance that each chosen label is correct.
- Which labelled step matches this detail? The missing result could change the all-ten correct count to 7 or 8.
- 1. Checked outcomes
- 2. Report coverage
- 3. Retain unknown
- 4. Keep scores
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Retain unknown The missing result could change the all-ten correct count to 7 or 8.

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Changed-case question

- Changed-case practice: The missing outcome is checked and found incorrect. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. All-ten agreement is 7/10=**70%**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: All-ten agreement is 7/10=**70%**. There are now three wrong labels and no unknown outcomes.

Repair and save your work

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

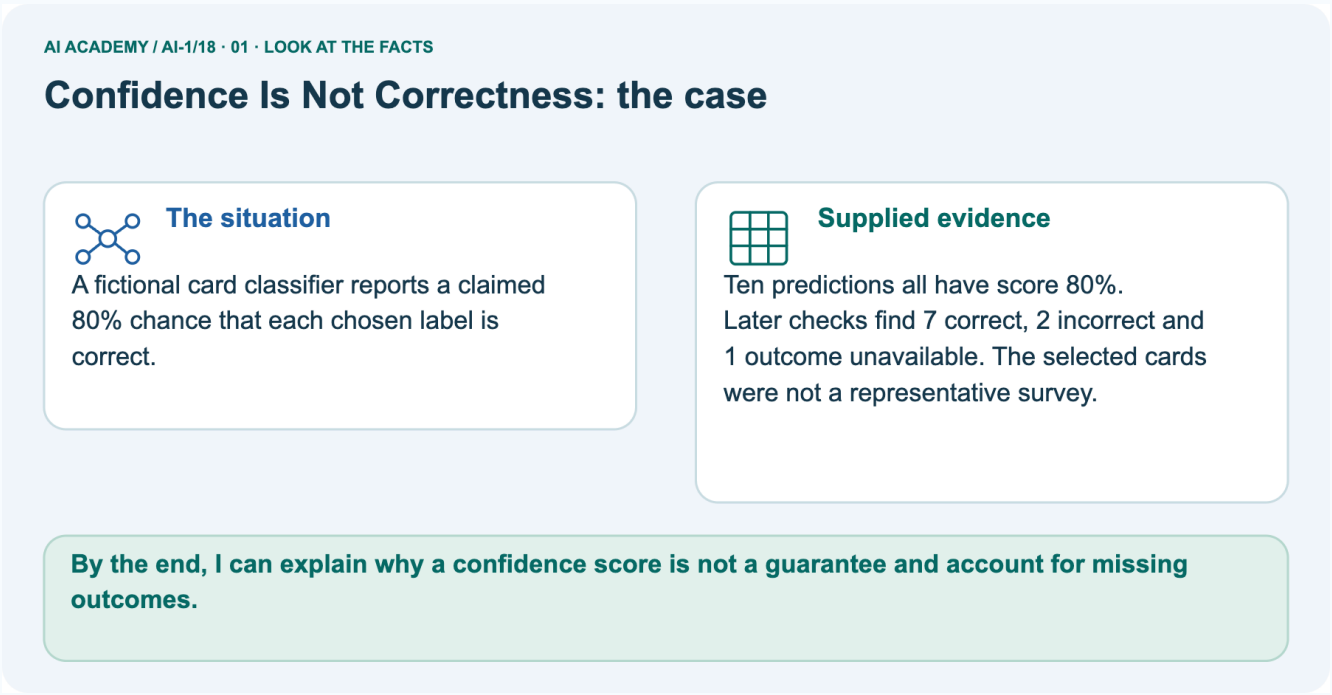
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Confidence Is Not Correctness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

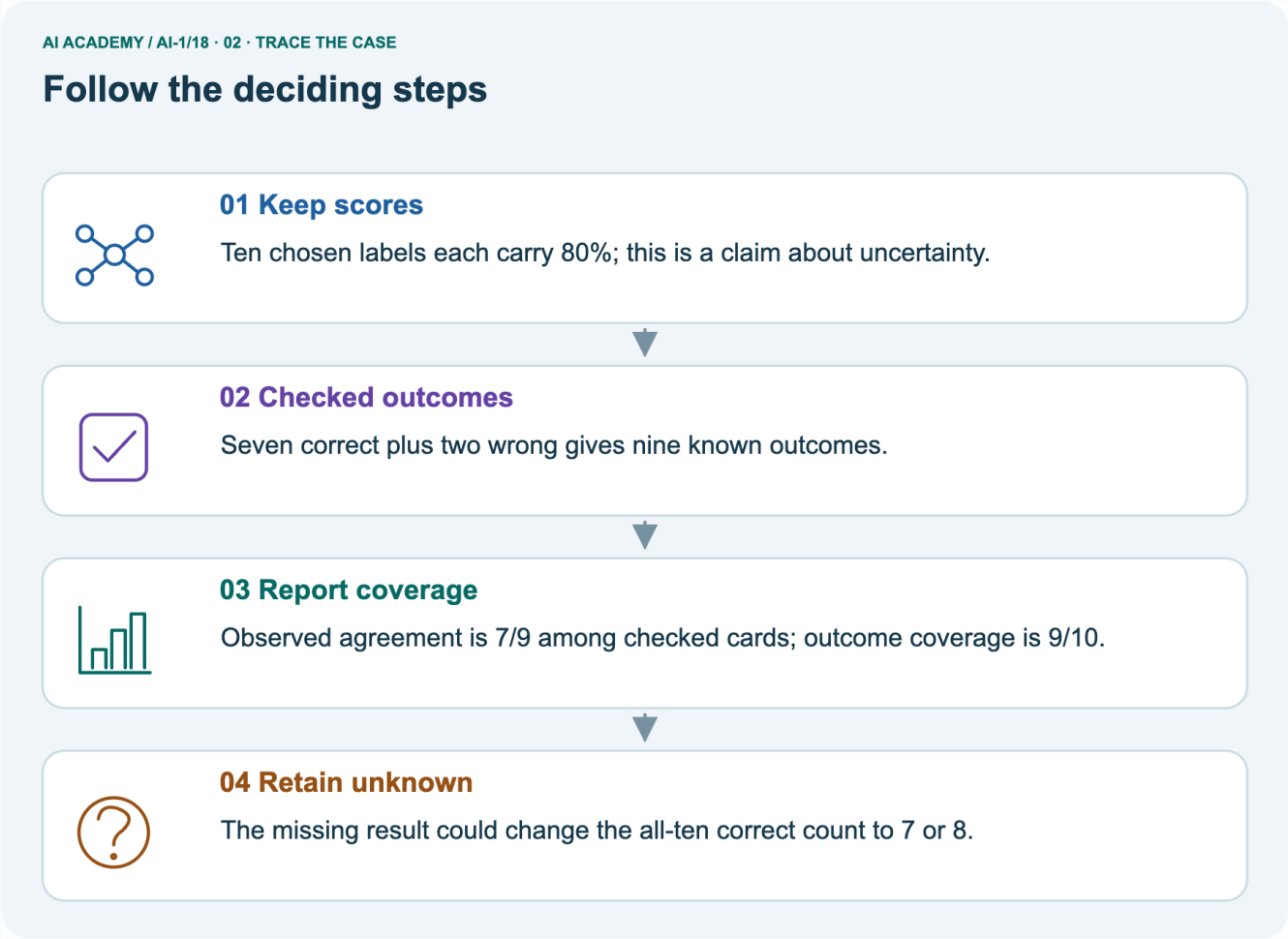
A fictional card classifier reports a claimed **80%** chance that each chosen label is correct. Supplied facts: Ten **prediction**s all have score **80%**. Later checks find 7 correct, 2 incorrect and 1 outcome unavailable.

The selected cards were not a representative survey.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep scores: Ten chosen labels each carry **80%**; this is a claim about uncertainty. → Checked outcomes: Seven correct plus two wrong gives nine known outcomes. → Report coverage: Observed agreement is 7/9 among checked cards; outcome coverage is 9/10. → Retain unknown: The missing result could change the all-ten correct count to 7 or 8.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Known outcomes and missing evidence



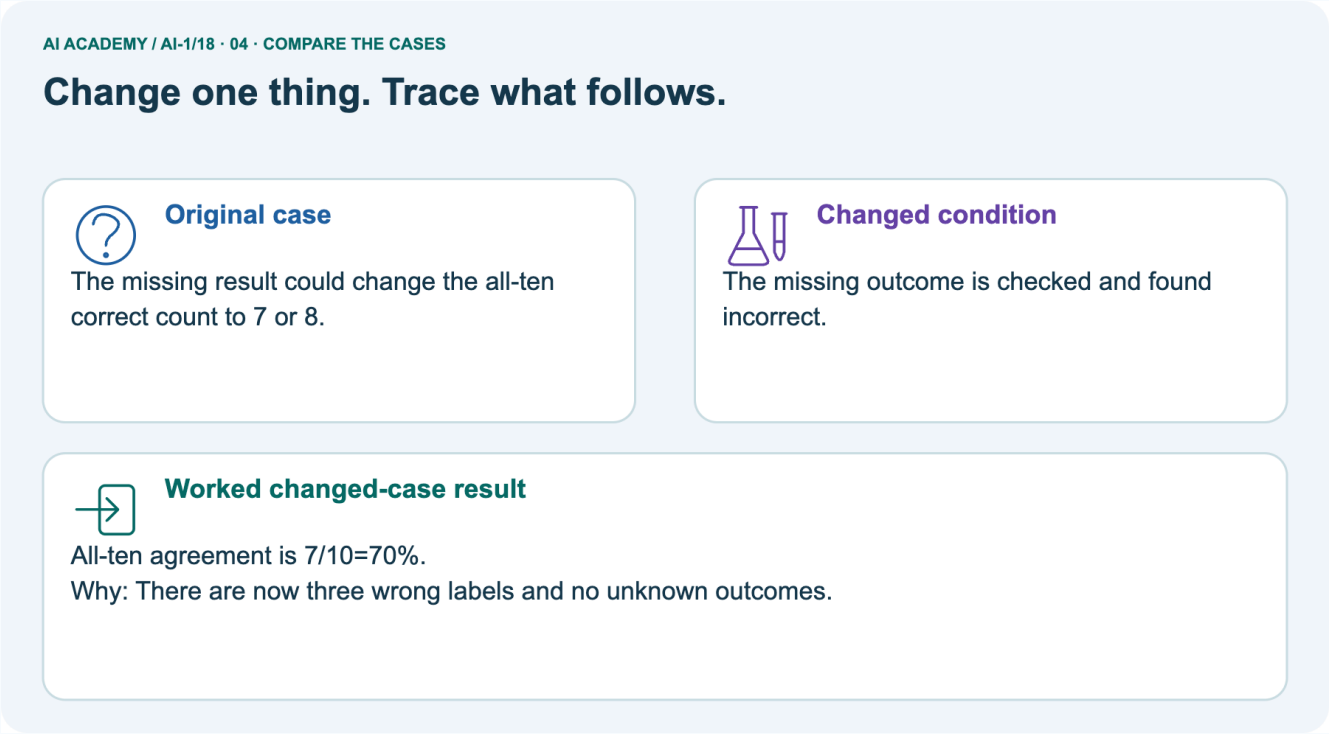
Follow the labelled picture with the matching explanation. Open the vector diagram

Known outcomes and missing **evidence**. Units: **prediction**s. Correct: 7. Incorrect: 2. Unknown: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

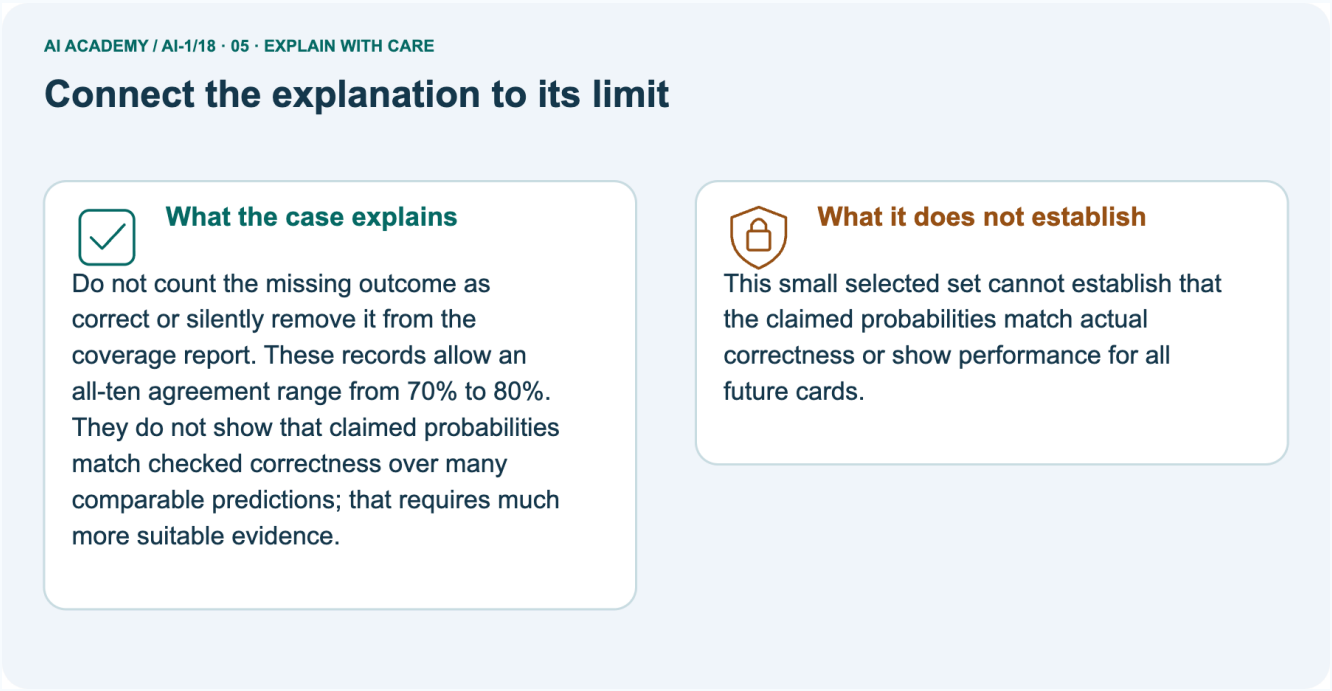
Original case: The missing result could change the all-ten correct count to 7 or 8. Changed condition: The missing outcome is checked and found incorrect. Expected result: All-ten agreement is 7/10=**70%**.

Reason: There are now three wrong labels and no unknown outcomes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Do not count the missing outcome as correct or silently remove it from the coverage report. These records allow an all-ten agreement range from **70%** to **80%**.

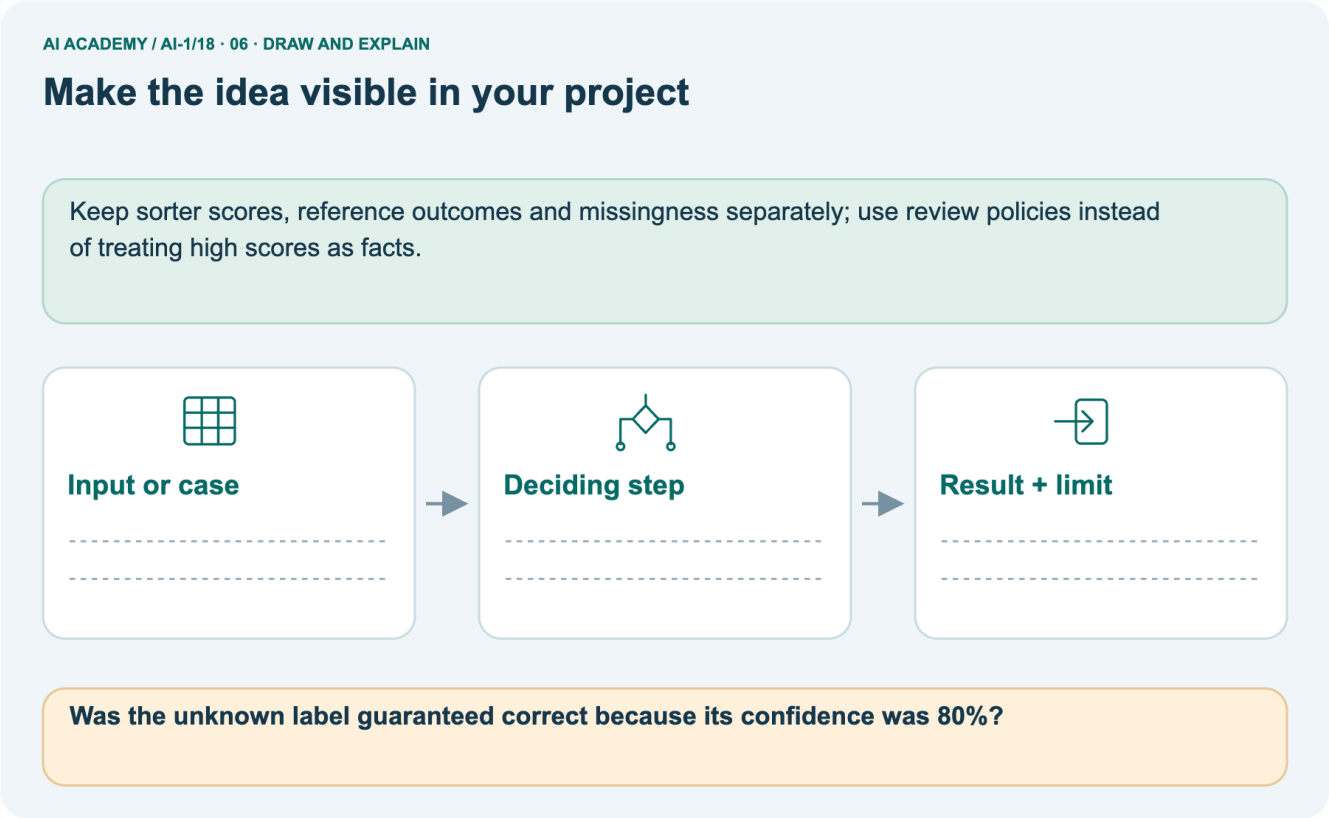
They do not show that claimed probabilities match checked **correctness** over many comparable **prediction**s; that requires much more suitable **evidence**.

Limit: This small selected set cannot establish that the claimed probabilities match actual **correctness** or show performance for all future cards.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep sorter scores, reference outcomes and missingness separately; use review policies instead of treating high scores as facts. Draw your **input**, deciding step, result and limit.

Supported question: Was the unknown label guaranteed correct because its **confidence** was **80%**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Confidence Is Not Correctness: the case

02. Follow the deciding steps

03. Known outcomes and missing evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.